Calcurion Maths Worksheets

A-Level Mathematics

Topic: Mechanics – Moments

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. A force of 10 N acts at a perpendicular distance of 0.5 m from a point. Find the moment. **[2]**

2. A force of 8 N acts at 30° to a rod of length 2 m, at one end. Find the moment about the other end. **[4]**

3. A uniform rod of length 4 m and mass 5 kg is pivoted at one end. Find the moment of its weight about the pivot. **[3]**

4. A uniform beam of length 6 m and mass 10 kg is supported at its ends. Find the reaction at each end. **[3]**

5. A uniform rod of length 2 m and weight 20 N is pivoted at one end and held horizontal by a vertical force at the other end. Find the force. **[3]**

6. A light rod of length 3 m has masses 2 kg and 4 kg at its ends. Find the position of the balance point. **[3]**

7. A uniform plank of length 5 m and mass 20 kg rests on two supports at its ends. A mass of 10 kg is placed 1 m from one end. Find the reactions at the supports. **[5]**

8. A force of 12 N acts perpendicular to a door of width 0.8 m. Find the moment about the hinges. **[2]**

9. A uniform rod AB of length 4 m and mass 6 kg is hinged at A and held horizontal by a string at B at 30° to the rod. Find the tension in the string. **[5]**

10. A particle of weight 10 N is suspended from a point by two strings of lengths 3 m and 4 m attached to points 5 m apart on a horizontal ceiling. Find the tensions. **[5]**

11. A uniform beam of mass 30 kg and length 4 m is supported by two vertical ropes at its ends. A mass of 20 kg is placed 1 m from one end. Find the tension in each rope. **[5]**

12. Find the moment of a 15 N force about a point 2 m away when the force is at 60° to the line joining the point to the point of application. **[3]**

13. A uniform rod of length 3 m and mass 4 kg is free to rotate about one end. It is held at 40° to the horizontal by a horizontal force at the other end. Find the force. **[5]**

14. A light rod of length 2 m carries a 5 kg mass at its midpoint. It is supported at both ends. Find the reactions. **[3]**

15. State the condition for a rigid body to be in equilibrium under a system of coplanar forces. **[2]**

Section B – Hard (Q16–25)

16. A uniform ladder of mass 15 kg and length 5 m rests against a smooth vertical wall with its base on rough horizontal ground, making 60° with the ground. Find the reactions at the wall and ground. **[6]**

17. A uniform beam AB of length 6 m and mass 20 kg is hinged at A and held horizontal by a cable from B to a point C which is 2 m vertically above A. Find the tension in the cable. **[6]**

18. A non-uniform rod of length 4 m has its centre of mass 1.5 m from one end. Its mass is 10 kg. It is supported at both ends. Find the reactions. **[5]**

19. A uniform ladder of mass 12 kg rests against a smooth wall, base on rough ground, at 50° to the ground. A person of mass 60 kg climbs one-third of the way up. Find the frictional force at the ground. **[7]**

20. A uniform rod of mass m and length $2a$ is freely hinged at one end and held at $30°$ to the horizontal by a string attached to the other end and to a point on the same level as the hinge. Find the tension. **[6]**

21. A light rod of length 5 m has masses 3 kg, 2 kg and 5 kg at distances 0, 2 m and 5 m from one end. Find the centre of mass. **[4]**

22. A uniform square lamina of side $2a$ and mass m is free to rotate about one side which is horizontal. It is held with its plane at $30°$ to the vertical by a horizontal force at the opposite side. Find the force. **[6]**

23. A see-saw of length 4 m and mass 20 kg is pivoted at its centre. A child of mass 30 kg sits 1.5 m from the centre. Where must a child of mass 25 kg sit to balance? **[4]**

24. A uniform rod AB of mass 8 kg and length 3 m is hinged at A. A force of 20 N acts vertically upwards at B. Find the reaction at the hinge when the rod is in equilibrium horizontal. **[5]**

25. A ladder of mass 20 kg and length 6 m rests against a smooth wall at $60°$ to the ground. The coefficient of friction at the ground is 0.4. Does the ladder slip? **[6]**

End of Worksheet • *Show all working clearly. Take* $g = 9.8\,\mathrm{m\,s^{-2}}$ *unless otherwise stated.*